

User Research Report: AI-Enabled Digital Platform for First-Time Mothers

Project Title: Intuitive Prenatal Care & Community Support
Research Methodology: Synthetic User Simulation (16 Profiles)

Project Team

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Initial Problem Statement

How might we design an AI-enabled platform that provides personalized health advice, pregnancy tracking, and peer support while navigating diverse cultural and socioeconomic contexts?

User Research Goal

Our goal is to understand the daily habits of first-time mothers how they track their pregnancy, learn about their health, and connect with other mothers online. We want to learn what makes them trust a digital tool so we can design a simple, easy-to-use platform that brings tracking, simple education, and safe community support into one place.

Research Methodology

Research Approach: Synthetic User Simulation (16 Profiles)

This research employs a synthetic user simulation methodology to generate diverse first-time mother personas across different geographic, economic, cultural, and social contexts. Rather than conducting traditional in-person interviews, we created detailed archetypal personas representing the spectrum of first-time mothers globally.

Data Collection Method:

- **Simulation-Based:** AI-generated personas based on real-world contexts and constraints
- **Contextual Diversity:** 16 personas spanning 8 countries across diverse economic tiers (Metro Tier 1, Tier 2, Tier 3, Semi-rural, Rural, Urban Slum)
- **Psychographic Depth:** Each persona includes personality traits, motivations, fears, attitudes, and behavioral patterns
- **Scenario-Based Analysis:** Personas evaluated against an 8-question questionnaire guide to identify pain points, gains, and platform design requirements

Why This Approach:

Synthetic personas allow rapid exploration of edge cases and diverse user contexts that would be expensive or logistically challenging to reach through traditional ethnographic research. This is particularly valuable for understanding pregnant mothers across the Global South, where access for research participation may be limited.

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Sampling Matrix

Context	Education Level	City	Key Psychographics	Profile
1	MBA	Mumbai, India	Highly organized, data-driven, slightly anxious	Priya Sharma
2	High School Diploma	Bhopal, India	Family-oriented, traditional, eager to learn	Sunita Devi
3	Currently in College	Chicago, USA	Expressive, socially conscious, easily overstimulated	Chloe Davis
4	Bachelor's in Graphic Design	London, UK	Extroverted, currently feeling isolated, visually oriented	Fatima Al-Hassan
5	Some College	Oaxaca, Mexico	Intuitive, nature-loving, cautious	Maria Hernandez
6	Vocational Training	Accra, Ghana	Resilient, practical, highly social	Grace Osei
7	Master's Degree	Tokyo, Japan	Private, meticulous, risk-averse	Yuki Takahashi
8	Bachelor's in Fine Arts	Berlin, Germany	Introverted, easily overwhelmed, detail-oriented (ADHD)	Nadia Kuznetsov
9	Bachelor's in Arts	Madurai, India	Respectful, family-oriented, slightly timid	Kavya Rangarajan
10	High School Dropout	São Paulo, Brazil	Independent, stressed, deeply loving	Bianca Silva
11	Bachelor of Technology (B.Tech)	Bengaluru, India	Optimization-focused, pragmatic, time-poor	Ananya Gowda
12	Bachelor of Commerce (B.Com)	Surat, India	Cheerful, business-minded, respectful of elders	Jignasha Patel
13	BSc in Nursing	Kochi, India	Clinically aware, lonely, highly communicative	Fatima Shajahan
14	12th Standard	Ludhiana District, India	Hardworking, stoic, community-driven	Manpreet Kaur
15	8th Grade Dropout	Dharavi, Mumbai, India	Street-smart, anxious about finances, protective	Rupali Mane
16	Bachelor of Education (B.Ed)	Guwahati, India	Inquisitive, gentle, focused on mental health	Meenakshi Borah

3. Target User Personas

To address the diversity found in the research, three distinct personas were developed to represent the spectrum of technological, cultural, and economic needs.

Persona 1: The Data-Driven Professional

"Knowledge is control. I need metrics, not metaphors."

- **Representative Users:** Priya (1), Yuki (7), Ananya Gowda (11), Fathima (13)
- **Background:** 32-year-old Senior IT Consultant in Bengaluru. Highly educated and tech-integrated.
- **Context:** Operates in a high-stress, high-performance environment. Relies on data to manage anxiety.
- **Goals:** Maintain professional reputation while ensuring a clinically perfect pregnancy.
- **Frustrations:** Dislikes "cute" or patronizing language; needs evidence-based, data-heavy explanations.

Daily Routine:

- **Morning:** Begins a high-pressure 10-hour shift at a tech firm, utilizing their commute to listen to medical podcasts or read peer-reviewed journals on a tablet.
- **Afternoon:** Spends the workday primarily sedentary at a desk, receiving automated health alerts from a smartwatch regarding resting heart rate spikes.
- **Evening:** Logs clinical data and ultrasound scans into private spreadsheets or shared portals to keep long-distance partners or medical providers informed.

Persona 2: The Cultural Navigator (Primary Persona)

"I want my doctor's safety and my family's blessing, but I often have to hide one from the other."

- **Representative Users:** Fatima Al-Hassan (4), Jignasha (12), Meenakshi Borah (16) (and Sunita (2), Maria Hernandez (5), Kavya (9))
- **Background:** 27-year-old Boutique Owner in Gujarat, living in a multi-generational household.
- **Context:** Caught in an "Information Tug-of-War" between medical advice and family traditions.
- **Goals:** Ensure baby's health without causing domestic conflict; maintain business autonomy.
- **Frustrations:** Feelings of loneliness due to privacy needs; difficulty discussing "taboo" physical changes.

Daily Routine:

- **Morning:** Manages the operations of a local boutique or business while navigating the domestic expectations of a multi-generational household.
- **Afternoon:** Carves out "stealth" windows to search for pregnancy symptoms in their local language, often locking themselves in a bedroom or office to hide their phone screen from judging elders.
- **Evening:** Participates in large family meals where they must carefully balance following medical prescriptions in secret while appearing to comply with traditional dietary advice from family.

Persona 3: The Resourceful Worker

"I need information that is fast, free, and speaks my language."

- **Representative Users:** Grace (6), Bianca (10), Manpreet (14), Rupali (15)

- **Background:** 24-year-old Domestic Worker in Mumbai or Farmer in Punjab.
- **Context:** High physical exertion jobs where missing a day of work means missing a day of pay.
- **Goals:** Ensure the baby is strong enough for birth; access government financial schemes (PMMVY).
- **Frustrations:** Medical jargon that hides costs; fear of being fired if physical limitations are noticed.

Daily Routine:

- **Morning:** Starts early with intense physical labor, such as cleaning high-rise apartments, working in agricultural fields, or standing for hours at a market stall.
 - **Afternoon:** Uses brief work breaks to send or receive voice notes on WhatsApp, preferring audio over text to save on mobile data and accommodate fast-paced environments.
 - **Evening:** Manages physical exhaustion and swelling while cross-referencing their next appointment date on a physical government hospital card or a kitchen wall calendar.
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4. Synthesis of Research Findings

A. Key Pains (The Emotional & Physical Toll)

- **Information Tug-of-War:** 80% of users suffer from “advice friction”—the stress of choosing between clinical prescriptions and cultural traditions.
- **The “Jargon Wall”:** Alienation caused by English-only medical terms, leading users to abandon searches.
- **Privacy Paranoia:** A constant need to hide digital activity (locking doors, private windows) to avoid family judgment.
- **Professional Fragility:** Fear of losing career momentum (High-income) or being fired (Low-income) due to pregnancy.

B. Key Gains (The Desired Outcomes)

- **2 AM Reassurance:** Immediate, non-judgmental AI support for late-night symptom anxiety.
- **The “Digital Bridge”:** Shared portals that help long-distance partners feel physically present.
- **Cultural Validation:** Feeling “seen” via local metaphors (fruits like mangoes/dates) and traditional practices.
- **Financial Navigation:** Discovering government schemes or cost-saving community tips.

C. Behavioural Patterns

- **Voice-First Navigation:** Heavy reliance on microphone buttons and voice notes due to literacy or labor constraints.
- **Visual Benchmarking:** Tracking progress through fetal-size imagery rather than technical text.
- **“Stealth” Usage Cycles:** Browsing in “safe windows” (commutes, bathrooms) to maintain privacy.
- **Algorithm over Anecdote:** Professional users prioritize automated dashboards over social forums.

D. Major Pain Points (Ranked by Criticality)

Rank	Pain Point	Impacted Users	Criticality
1	Domestic Conflict	2, 4, 5, 9, 12, 14	High: Users will abandon apps that cause family friction.
2	Language Barrier	2, 6, 9, 14, 15	Critical: Medical English is a hard barrier for the most vulnerable.
3	Data Privacy	1, 7, 11, 13	High: Risk of professional reputation damage or insurance leaks.
4	Financial Anxiety	6, 10, 15	Moderate: Health tracking must include work-capacity advice.

5. Actionable Feature Recommendations

Feature 1: The "Tradition-Science Bridge" (AI Explainer)

- **Insight:** Users feel torn between elders and doctors.
- **Solution:** An AI module that accepts traditional advice (e.g., "Don't eat papaya") and provides a respectful, science-based "middle ground" explanation to help mothers negotiate with family.

Feature 2: "Stealth Mode" UI & Privacy Vault

- **Insight:** Users hide their screens to avoid judgment.
- **Solution:** A camouflage app icon (looking like a Calculator or Notes app) with a PIN-protected vault for journals and a "Quick-Exit" button to hide the screen instantly.

Feature 3: Voice-First "ASHA" Interface

- **Insight:** Users trust oral advice over "Big Words."
- **Solution:** An audio-centric UI where the AI acts as a "Digital ASHA worker," allowing users to ask questions and receive answers in local dialects (Tamil, Punjabi, Marathi).

Feature 4: Shared "Partner Portal"

- **Insight:** Long-distance partners feel disconnected from the pregnancy.
- **Solution:** A dual-access dashboard where partners can view growth trends, scans, and "To-Do" lists to foster bonding across distances.

Feature 5: Work-Capacity & Benefit Tracker

- **Insight:** Low-income users prioritize job security and government aid.
 - **Solution:** A tracker for the physical demands of labor-intensive jobs and an integrated guide for local government financial schemes (e.g., PMMVY).
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Feature Prioritization

Priority Matrix: Impact vs. Feasibility

Priority	Feature	Impact	Feasibility	Rationale
1 (High)	Stealth Mode UI & Privacy Vault	Critical	Medium	Addresses the #1 pain point (Domestic Conflict). Users will abandon apps that expose their activity to family judgment. Privacy is non-negotiable.
2 (High)	Tradition-Science Bridge (AI Explainer)	Critical	High	Directly reduces Information Tug-of-War (80% of users). Enables mothers to negotiate between elders and doctors without app abandonment.
3 (High)	Voice-First "ASHA" Interface	Critical	Medium	Solves Language Barrier (the #2 most critical pain point). Essential for the most vulnerable users (6 personas: Sunita, Grace, Kavya, Manpreet, Rupali, others).
4 (Medium)	Shared "Partner Portal"	High	Medium	Addresses emotional isolation and strengthens family engagement. Benefits 4 personas (Priya, Ananya, Fatima, Fathima).
5 (Medium)	Work-Capacity & Benefit Tracker	High	Low	Impacts low-income users (4 personas: Grace, Bianca, Manpreet, Rupali) but requires significant backend integration with government databases and labor tracking.

Limitations & Reflection

Evolution of Insights

Research began with a focus on Technical UX (making charts clearer). As the pool grew to 16, the focus shifted to **Agency and Power Dynamics**, revealing that information is useless if a mother lacks the domestic authority to act on it. Finally, the **Economic Layer** emerged, showing that pregnancy is often managed as a financial risk as much as a health event.

Key Limitations of This Research

- **Narrative "Cleanliness":** Synthetic users are highly articulate; real users often struggle to voice fears or say "I don't know."
 - **Lack of Environmental Context:** AI provides transcripts but misses non-verbal cues (hesitation, physical discomfort, or the visual environment).
 - **Language Nuance:** While AI can simulate local preferences, it often misses the specific "code-switching" and regional slang of local communities.
 - **Idealized Trust:** Synthetic users are generally more willing to trust AI than real-world users, who may harbor deep skepticism of digital machines.
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8. User Interview Log (16 Participants)

Questionnaire Guide

1. Walk me through the last time you had a question about a body change. What did you do first?
2. Tell me about the apps, websites, or people you use to learn about your pregnancy.
3. What do you do when you see medical words you do not know?
4. Show me how you track your pregnancy progress.
5. Describe your experience using AI tools like ChatGPT or Gemini for pregnancy questions.
6. How often do you feel torn between your doctor's advice and your family's advice?
7. What is happening to your body right now that no one has explained well?
8. What things are you feeling that you choose not to share with others?

Participant Reference (Number to Name)

1. Priya
2. Sunita
3. Chloe
4. Fatima Al-Hassan
5. Maria Hernandez
6. Grace
7. Yuki
8. Nadia
9. Kavya
10. Bianca
11. Ananya Gowda
12. Jignasha
13. Fathima
14. Manpreet
15. Rupali
16. Meenakshi Borah

Persona Description Document

[Persona Description \(GitHub PDF\)](#)

[Persona Description \(Google Drive\)](#)

Interview Transcripts Document

[Interview Transcripts \(GitHub PDF\)](#)

[Interview Transcripts \(Google Drive\)](#)